

AI Fundamentals

A beginner-friendly knowledge document for testing a Retrieval-Augmented Generation (RAG) Agent

1. Artificial Intelligence

Artificial Intelligence, or AI, is the field of building computer systems that can perform tasks that normally require human intelligence. Examples include understanding language, recognizing images, making predictions, planning, and solving problems. AI systems can use rules, machine learning models, or combinations of different techniques.

2. Machine Learning

Machine Learning is a part of AI in which a system learns patterns from data instead of being programmed with a separate rule for every situation. Common learning types include supervised learning, unsupervised learning, and reinforcement learning. A trained model can use learned patterns to make predictions or decisions on new data.

3. Deep Learning

Deep Learning is a branch of machine learning that uses neural networks with multiple layers. Deep learning is widely used for image recognition, speech processing, natural language processing, recommendation systems, and generative AI.

4. Large Language Models

A Large Language Model, or LLM, is a neural network trained on large amounts of text and other data so it can understand and generate language. LLMs can answer questions, summarize information, write text, translate languages, and assist with programming. An LLM by itself does not automatically know private documents or live web information.

5. Generative AI

Generative AI refers to systems that create new content such as text, images, audio, video, or code. Modern language models are examples of generative AI because they generate responses based on a user's prompt and the context available to them.

6. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with text generation. Instead of relying only on the model's learned knowledge, a RAG system first searches a

knowledge source for relevant information and then provides that information to the language model. A typical RAG pipeline includes document loading, text cleaning, chunking, embeddings, vector storage, retrieval, and answer generation.

7. Embeddings

An embedding converts text or another piece of information into a numerical vector that represents its semantic meaning. Similar pieces of information tend to have vectors that are close to one another in vector space. Embeddings make semantic search possible.

8. Vector Databases

A vector database stores embeddings and supports similarity search. In a RAG system, document chunks are converted into embeddings and stored in a vector database. When a user asks a question, the question is also embedded and the database retrieves chunks with similar meaning. ChromaDB is one example of a vector database used in AI applications.

9. AI Agents

An AI Agent is a software system that can interpret a user's goal, decide what action is needed, use available tools, observe the results, and produce an answer. Tools can include calculators, web search, databases, APIs, code execution, and document retrieval. An agent is therefore more action-oriented than a basic chatbot.

10. AI Agent vs Chatbot

A basic chatbot mainly receives a message and generates a response. An AI Agent can additionally choose and use tools to accomplish tasks. For example, a chatbot may explain what 25 multiplied by 50 means, while an agent can call a calculator tool and return the computed result. Agents can also combine tools, memory, web search, and retrieval.

11. Persistent Memory

Persistent memory allows an AI application to retain information between program runs. A database such as SQLite can store previous user and assistant messages. When the application starts again, stored information can be loaded and supplied as context when appropriate.

12. Ollama and Local Models

Ollama provides a local runtime for running supported AI models on a user's computer. In a local AI project, Ollama can run a model such as Qwen3:4b and expose it to applications through a local service. This approach can reduce dependence on paid cloud model APIs and keeps model execution on the local machine, subject to the computer's available resources.

13. LangChain

LangChain is a framework for building applications around language models. It provides components for models, prompts, tools, agents, retrieval, and integrations. In an agent project, LangChain can connect a local language model with tools such as a calculator, web search, and a RAG retriever.

14. Example End-to-End AI Agent

A practical local AI Agent can combine several capabilities. A user asks a question. The agent interprets the request and chooses whether to answer directly or call a tool. For a calculation it uses a calculator. For current information it uses web search. For information stored in a PDF it uses RAG. For previous conversations it can use persistent memory. The selected information is then provided to the language model, which generates the final response.

15. Key Takeaways

AI is the broad field. Machine Learning is one approach within AI, and Deep Learning is a machine learning approach based on neural networks. LLMs specialize in language generation and understanding. RAG gives an LLM access to external knowledge. AI Agents add decision-making and tool use. Embeddings enable semantic search, vector databases store embeddings, and persistent memory allows useful information to survive between sessions.